

Classical Electrodynamics

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Abstract

1 Old Math

1.1 Vectors

$$a \cdot (b \times c) = b \cdot (c \times a) = c \cdot (a \times b) \quad (1)$$

$$a \times (b \times c) = (a \cdot c)b - (a \cdot b)c \quad (2)$$

$$(a \times b) \cdot (c \times d) = (a \cdot c)(b \cdot d) - (a \cdot d)(b \cdot c) \quad (3)$$

$$\nabla \times \nabla \psi = 0 \quad (4)$$

$$\nabla \cdot (\nabla \times a) = 0 \quad (5)$$

$$\nabla \times (\nabla \times a) = \nabla(\nabla \cdot a) - \nabla^2 a \quad (6)$$

$$\nabla \cdot (\psi a) = a \cdot \nabla \psi + \psi \nabla \cdot a \quad (7)$$

$$\nabla \times (\psi a) = \nabla \psi \times a + \psi \nabla \times a \quad (8)$$

$$\nabla(a \cdot b) = (\nabla a) \cdot b + (b \cdot \nabla)a + a \times (\nabla \times b) + b \times (\nabla \times a) \quad (9)$$

$$\nabla \cdot (a \times b) = b \cdot (\nabla \times a) - a \cdot (\nabla \times b) \quad (10)$$

$$\nabla \times (a \times b) = a(\nabla \cdot b) - b(\nabla \cdot a) + (b \cdot \nabla)a - (a \cdot \nabla)b \quad (11)$$

Divergence Theorem

$$\int_V \nabla \cdot A d^3x = \int_S A \cdot n da \quad (12)$$

$$\int_V \nabla \psi d^3x = \int_S \psi n da \quad (13)$$

$$\int_V \nabla \times A d^3x = \int_S n \times A da \quad (14)$$

Green's first Identity

$$\int_V (\phi \nabla^2 \psi + \nabla \phi \cdot \nabla \psi) d^3x = \int_S \phi \mathbf{n} \cdot \nabla \psi da \quad (15)$$

Greens Theorem

$$\int_V (\phi \nabla^2 \psi - \psi \nabla^2 \phi) d^3x = \int_S (\phi \nabla \psi - \psi \nabla \phi) \cdot \mathbf{n} da \quad (16)$$

Stoke's Theorem

$$\int_S (\nabla \times \mathbf{A}) \cdot d\mathbf{a} = \oint_C \mathbf{A} \cdot d\mathbf{l} \quad (17)$$

$$\int_S \mathbf{n} \times \nabla \psi da = \oint_C \psi d\mathbf{l} \quad (18)$$

C is the contour bounding the Open Surface S with $d\mathbf{l}$ as the line element of the contour

Vector Formulas

$$\begin{aligned} \mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) &= \mathbf{b} \cdot (\mathbf{c} \times \mathbf{a}) = \mathbf{c} \cdot (\mathbf{a} \times \mathbf{b}) \\ \mathbf{a} \times (\mathbf{b} \times \mathbf{c}) &= (\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{a} \cdot \mathbf{b})\mathbf{c} \\ (\mathbf{a} \times \mathbf{b}) \cdot (\mathbf{c} \times \mathbf{d}) &= (\mathbf{a} \cdot \mathbf{c})(\mathbf{b} \cdot \mathbf{d}) - (\mathbf{a} \cdot \mathbf{d})(\mathbf{b} \cdot \mathbf{c}) \\ \nabla \times \nabla \psi &= 0 \\ \nabla \cdot (\nabla \times \mathbf{a}) &= 0 \\ \nabla \times (\nabla \times \mathbf{a}) &= \nabla(\nabla \cdot \mathbf{a}) - \nabla^2 \mathbf{a} \\ \nabla \cdot (\psi \mathbf{a}) &= \mathbf{a} \cdot \nabla \psi + \psi \nabla \cdot \mathbf{a} \\ \nabla \times (\psi \mathbf{a}) &= \nabla \psi \times \mathbf{a} + \psi \nabla \times \mathbf{a} \\ \nabla(\mathbf{a} \cdot \mathbf{b}) &= (\mathbf{a} \cdot \nabla)\mathbf{b} + (\mathbf{b} \cdot \nabla)\mathbf{a} + \mathbf{a} \times (\nabla \times \mathbf{b}) + \mathbf{b} \times (\nabla \times \mathbf{a}) \\ \nabla \cdot (\mathbf{a} \times \mathbf{b}) &= \mathbf{b} \cdot (\nabla \times \mathbf{a}) - \mathbf{a} \cdot (\nabla \times \mathbf{b}) \\ \nabla \times (\mathbf{a} \times \mathbf{b}) &= \mathbf{a}(\nabla \cdot \mathbf{b}) - \mathbf{b}(\nabla \cdot \mathbf{a}) + (\mathbf{b} \cdot \nabla)\mathbf{a} - (\mathbf{a} \cdot \nabla)\mathbf{b} \end{aligned}$$

If $\mathbf{x}$ is the coordinate of a point with respect to some origin, with magnitude $r = |\mathbf{x}|$, $\mathbf{n} = \mathbf{x}/r$ is a unit radial vector, and $f(r)$ is a well-behaved function of r , then

$$\begin{aligned} \nabla \cdot \mathbf{x} &= 3 & \nabla \times \mathbf{x} &= 0 \\ \nabla \cdot [\mathbf{n}f(r)] &= \frac{2}{r}f + \frac{\partial f}{\partial r} & \nabla \times [\mathbf{n}f(r)] &= 0 \\ (\mathbf{a} \cdot \nabla)\mathbf{n}f(r) &= \frac{f(r)}{r}[\mathbf{a} - \mathbf{n}(\mathbf{a} \cdot \mathbf{n})] + \mathbf{n}(\mathbf{a} \cdot \mathbf{n})\frac{\partial f}{\partial r} \\ \nabla(\mathbf{x} \cdot \mathbf{a}) &= \mathbf{a} + \mathbf{x}(\nabla \cdot \mathbf{a}) + i(\mathbf{L} \times \mathbf{a}) \end{aligned}$$

where $\mathbf{L} = \frac{1}{i}(\mathbf{x} \times \nabla)$ is the angular-momentum operator.

(a) Vector Formulas

Theorems from Vector Calculus

In the following ϕ , ψ , and $\mathbf{A}$ are well-behaved scalar or vector functions, V is a three-dimensional volume with volume element d^3x , S is a closed two-dimensional surface bounding V , with area element da and unit outward normal $\mathbf{n}$ at da .

$$\int_V \nabla \cdot \mathbf{A} d^3x = \int_S \mathbf{A} \cdot \mathbf{n} da \quad (\text{Divergence theorem})$$

$$\int_V \nabla \psi d^3x = \int_S \psi \mathbf{n} da$$

$$\int_V \nabla \times \mathbf{A} d^3x = \int_S \mathbf{n} \times \mathbf{A} da$$

$$\int_V (\phi \nabla^2 \psi + \nabla \phi \cdot \nabla \psi) d^3x = \int_S \phi \mathbf{n} \cdot \nabla \psi da \quad (\text{Green's first identity})$$

$$\int_V (\phi \nabla^2 \psi - \psi \nabla^2 \phi) d^3x = \int_S (\phi \nabla \psi - \psi \nabla \phi) \cdot \mathbf{n} da \quad (\text{Green's theorem})$$

In the following S is an open surface and C is the contour bounding it, with line element $d\mathbf{l}$. The normal $\mathbf{n}$ to S is defined by the right-hand-screw rule in relation to the sense of the line integral around C .

$$\int_S (\nabla \times \mathbf{A}) \cdot \mathbf{n} da = \oint_C \mathbf{A} \cdot d\mathbf{l} \quad (\text{Stokes's theorem})$$

$$\int_S \mathbf{n} \times \nabla \psi da = \oint_C \psi d\mathbf{l}$$

(b) Vector Theorems

$$\hat{r} = \frac{\vec{r}}{|\vec{r}|}$$

On page 1 equation 1

1.2 Dirac Delta Properties

Where the prime is differentiation with respect to the argument.

$$\delta(x - a) = a \text{ for } x \neq a \quad (19)$$

if the region of integration includes $x=a$

$$\int \delta(x - a) dx = 1 \quad (20)$$

and is zero otherwise ($x \neq a$)

$$\int \delta(x - a) dx = 0 \quad (21)$$

$$\int f(x)\delta(x-a)dx = f(a) \quad (22)$$

$$\int f(x)\delta'(x-a)dx = -f'(a) \quad (23)$$

$$\delta(f(x)) = \sum \frac{1}{\frac{df}{dx}x_i} \delta(x-x_i) \quad (24)$$

$$\delta^{(3)} = U\delta(x-x_o)V\delta(y-y_0)W\delta(z-z_0) \quad (25)$$

If ΔV contains $x = X$

$$\int_{\Delta V} \delta(x-X)d^3x = 1 \quad (26)$$

If ΔV does not contain $x = X$

$$\int_{\Delta V} \delta(x-X)d^3x = 0 \quad (27)$$

$$\delta(au) = \frac{1}{|a|}\delta(u) \quad (28)$$

1.3 Various Old Math

The 3D volume element at dx'

$$d^3x' = dx'dy'dz' \quad (29)$$

2 New Math

$$\rho_q = q_i\delta^{(3)}(x-x_i) \quad (30)$$

$$\rho_\lambda = \lambda_i\delta^{(2)} \quad (31)$$

$$\rho_\sigma = \sigma_i\delta^{(1)} \quad (32)$$

3 Electrodynamics

3.1 Putting charge at origin

Using general 3D Dirac Delta function (Reference equ 25 on pg 3)

$$\delta^{(3)} = U\delta(x-x_o)V\delta(y-y_0)W\delta(z-z_0)$$

Replace Cartesian with spherical coordinates (r, θ, ϕ)

Modify U, V, W with incremental length elements $(dr, r d\theta, r \sin(\theta) d\phi)$

This makes $U = 1, V = \frac{1}{r}, W = \frac{1}{r \sin(\theta)}$

$$\delta^{(3)} = \delta(r-r_0) \frac{\delta(\theta-\theta_0)}{r} \frac{\delta(\phi-\phi_0)}{r \sin(\theta)} \quad (33)$$

For 2D data in spherical coordinates we consider a circular 2D ring with

a radius of R , lying in the x-y plane. This makes $r_0 = R$, $\theta_0 = \frac{\pi}{2}$, and $\phi - \phi_0 = 0$

$$\delta_{ring}^{(2)} = \delta(r - R) \frac{\delta(\theta - \frac{\pi}{2})}{r} \quad (34)$$

For thin spherical shell. This makes $r_0 = R$, $\theta - \theta_0 = 0$, and $\phi - \phi_0 = 0$

$$\delta_{shell}^{(1)} = \delta(r - R) \quad (35)$$

For point charge at origin in spherical coordinate

$$\delta_{origin}^{(3)} = \frac{\delta(r)}{4\pi r^2} \quad (36)$$

$$\vec{E} = k \frac{q}{r^2} \hat{r} \quad (37)$$

For line charge along z axis of cylindrical coordinates

$$\delta_{cylLine}^{(2)} = \frac{\delta(\rho)}{2\pi\rho} \quad (38)$$

3.2 Coulomb's Law

Two Electric point charges exert a force on each other.

$$\vec{F} = q\vec{E} = kq_1q_2 \frac{x_1 - x_2}{|x_1 - x_2|^3} \quad (39)$$

Electric field at point x due to a point charge

$$\vec{E} = kq_1 \frac{x - x_1}{|x - x_1|^3} \quad (40)$$

In electrostatic units (esu) $k = 1$, In SI:

$$k = \frac{1}{4\pi\epsilon_0} = 10^{-7}c^2 \quad (41)$$

Since Electrodynamics is a macroscopic theory, we will consider the "point" to be many charged sources. Experiment shows that the fields add linearly.

$$\vec{E}(x) = k \sum_i q_i \frac{x - x_i}{|x - x_i|^3} \quad (42)$$

Consider the many charged sources to be "small" (Replace q with ρ) and "numerous" (Replace Sum with Integral) at point x' . Coulomb's law in terms

of the field for a charge density. Replace singular x dimension with 3D x,y,z dimension (Reference 29 on pg 3)

$$\vec{E}(x) = k \int \rho_q(x') \frac{x - x'}{|x - x'|^3} d^3x' \quad (43)$$

Replacing point charge density with sheet charge density (Reference 31 on pg 3) we get coulomb's integral form for the electric field at point x due to a sheet charge

$$\vec{E}(x) = k \int \sigma(x') \frac{x - x'}{|x - x'|^3} d^3x' \quad (44)$$

Replacing point charge density with line charge density (Reference 32 on pg 3) we get coulomb's integral form for the electric field at point x due to a line charge

$$\vec{E}(x) = k \int \lambda(x') \frac{x - x'}{|x - x'|^3} d^3x' \quad (45)$$

The following is proof of the previous formulation

Describe charge density using Dirac Delta function properties (Reference 30 on pg 3)

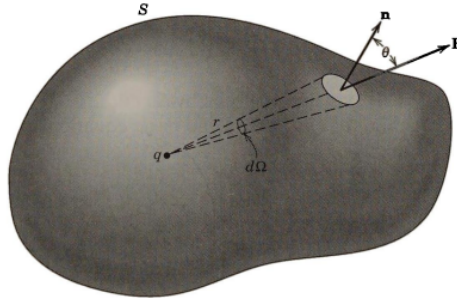
$$\vec{E}(x) = k \int \sum q_i \delta(x - x_i) \frac{x - x'}{|x - x'|^3} d^3x' \quad (46)$$

After Integration this verifies our earlier assumption (Reference equ 40 on pg 4)

$$\vec{E} = kq_1 \frac{x - x_1}{|x - x_1|^3}$$

Coulomb's Law does not include bounding surfaces. Therefore the integral must be done over all charges in the universe, or we must assume that most of the charges in the universe are far enough away that their contribution is negligible. Because of this fact, Coulomb's law has limited applicability.

3.3 Gauss Law



(c) Gauss Law Formula Variables

Take Coulomb's law for a point charge (Ref equ 40 on pg 4)

$$\vec{E}(x) = kq \frac{x - x_1}{|x - x_1|^3}$$

As before, put the charge at the origin (Ref equ 37 on pg 4)

$$\vec{E} = k \frac{q}{r^2} \hat{r}$$

Dot both sides by the normal vector

$$\vec{E} = k \frac{q}{r^2} \cos(\theta) \quad (47)$$

Integrate both sides over the closed surface

$$\oint_S (\vec{E} \cdot n) da = \oint_S (k \frac{q}{r^2} \cos(\theta)) da \quad (48)$$

Since E is directed along the line from surface element to the charge q

$$\oint_S (\vec{E} \cdot n) da = \oint_S (k \frac{q}{r^2} r^2) d\Omega \quad (49)$$

Simple algebra eliminates r^2 , and take constants out of integral

$$\oint_S (\vec{E} \cdot n) da = kq \oint_S d\Omega \quad (50)$$

Expand k (Ref equ 41 on pg 4)

$$\oint_S (\vec{E} \cdot n) da = \frac{q}{4\pi\epsilon_0} \oint_S d\Omega \quad (51)$$

Integrate Θ to get total charge enclosed by Gaussian surface.

$$\oint_S d\Theta = \int_0^{2\pi} \int_0^\pi d\Theta \quad (52)$$

Replace Θ with cylindrical coordinates (Ref fig 1e on pg 1e

$$\oint_S d\Theta = \int_0^{2\pi} \int_0^\pi r^2 \sin(\theta) d\theta d\phi = 4\pi \quad (53)$$

Place Integration back into equation (Ref equ ?? on pg ??)

$$\oint_S (\vec{E} \cdot n) da = \frac{4q\pi}{4\pi\epsilon_0} = \frac{q}{\epsilon_0} \quad (54)$$

The above equation is true for any charge completely enclosed by Gaussian surface. For any charge outside of Gaussian surface:

$$\oint_S \vec{E} \cdot n da = 0 \quad (55)$$

For a discrete set of charges

$$\oint_S \vec{E} \cdot n da = \frac{\sum q_i}{\epsilon_0} \quad (56)$$

For continuous charge density

$$\boxed{\oint_S \vec{E} \cdot n da = \frac{\int_V \rho_q(x) d^3x}{\epsilon_0}} \quad (57)$$

Apply Divergence Theorem (Ref equ 12 on pg 1)

$$\int_V \nabla \cdot \vec{E} d^3x = \frac{\int_V \rho_q(x) d^3x}{\epsilon_0} \quad (58)$$

Apply simple algebra to get Gauss's Law in differential form

$$\int_V (\nabla \cdot \vec{E} - \frac{\rho_q(x)}{\epsilon_0}) d^3x = 0 \quad (59)$$

$$\boxed{\nabla \cdot \vec{E} = \frac{\rho_q(x)}{\epsilon_0}} \quad (60)$$

In comparison to Coulomb's Law and Gauss's law in integral form, Gauss's differential equation is local and therefore most useful in practice.

For the very specific case of the electric field being constant across the surface and everywhere perpendicular to the surface

$$\vec{E} = E_0 n \quad (61)$$

$$E_0 \oint_S da = \frac{\int_V \rho_q(x) d^3x}{\epsilon_0} \quad (62)$$

$$E_0 A_S = \frac{\int_V \rho_q(x) d^3x}{\epsilon_0} \quad (63)$$

$$E_0 = \frac{\int_V \rho_q(x) d^3x}{\epsilon_0 A_S} \quad (64)$$

3.4 Scalar Potential

A vector field can be specified completely if its divergence and curl are given everywhere in space, except up to the gradient of scalar function that satisfies the Laplace equation.

Therefore we need to specify the curl of $\vec{E}$

Starting at Coulomb's generalized integral form (Reference equation 43 on page 5

$$\vec{E}(x) = k \int \rho_q(x') \frac{x - x'}{|x - x'|^3} d^3x'$$

Using Property of Nabla (∇)

$$-\nabla\left(\frac{1}{|x - x'|}\right) = \frac{x - x'}{|x - x'|^3} \quad (65)$$

Substitute equation 65 into 43

$$\vec{E}(x) = k \int \rho_q(x') -\nabla\left(\frac{1}{|x - x'|}\right) d^3x' \quad (66)$$

Nabla (∇) is constant; therefore,

$$\vec{E}(x) = -k\nabla \int \rho_q(x') \left(\frac{1}{|x - x'|}\right) d^3x' \quad (67)$$

Cross nabla (∇) on both sides of equation and substitute equation 4

$$\nabla \times \vec{E}(x) = \nabla \times k - k\nabla \int \rho_q(x') \left(\frac{1}{|x - x'|}\right) d^3x' \quad (68)$$

Therefore,

$$\nabla \times \vec{E} = 0 \quad (69)$$

By manipulating equation 67

$$\vec{E}(x) = \nabla \int \left(\frac{k\rho_q(x')}{|x - x'|}\right) d^3x' \quad (70)$$

We can now define a portion of the equation as the scalar potential (Φ)

$$\vec{E}(x) = \nabla\Phi \quad (71)$$

Work done can be defined as moving a charge from point A to point B in an electric field

$$W = - \int_A^B \vec{F} \cdot dx \quad (72)$$

Substitute equation 39 into

$$W = -q \int_A^B \vec{E} \cdot dx$$

Substitute from previously formulated equ

$$W = q \int_A^B \nabla \Phi \cdot dx$$

$$W = q \int_A^B d\Phi$$

Integrate

$$W = q(\Phi_B - \Phi_A)$$

By work-energy theorem, work and energy are equivalent, therefore

$$\Delta \Phi = \frac{U}{q}$$

Substitute previously formulated

$$-q \int_A^B \vec{E} \cdot dx = q(\Phi_B - \Phi_A)$$

if path is closed

$$\oint \vec{E} \cdot dx = 0$$

3.5 Capacitance

Collection of separate objects that are perfect conductors residing in free space. The charge, conductor potentials and geometric properties are the only links to the free space and the conductors.

The electrostatic potential at any point in space due to a collection of n objects is just the sum of all individual potentials

$$\Phi = k \sum \int \frac{\rho_j(x')}{|x-x'|} dx'$$

We only care about points in space on the surface of the objects, so x_i is a point on the surface of i object

$$\Phi(x_i) = k \sum \int \frac{\rho_j(x')}{|x_i-x'|} dx'$$

Since normalized charge density $\frac{\rho_j(x')}{Q_j}$ is purely geometrical, where Q_j is the total charge on the plate we can do the following. This is an approximation/assumption because the charge density "tends" to spread out into the same minimal energy pattern when a charge is applied.

$$\Phi(x_i) = k \sum_j \int \frac{\rho_j(x') Q_j}{|x_i-x'| Q_j} dx'$$

Simple algebra and removing Q_j from the integral

$$\Phi(x_i) = \sum Q_j k \int \frac{\rho_j(x')}{|x_i-x'|} dx'$$

We now define the Geometric Coefficient p_{ij} which is the normalized electric potential felt at the i^{th} conductor due to the j^{th} conductor.

$$p_{ij} = k \int \frac{\rho_j(x')}{|x_i-x'|} dx'$$

$$\Phi(x_i) = \sum p_{ij} Q_j$$

Invert equation for more practical use

$$Q_i = \sum p_{ij} \Phi_j$$

We now define the generalized Geometric Coefficient as Capacitance C

$$Q_i = \sum C_{ij} \Phi_j$$

For 2 conductors, we have

$$Q_1 = C_{11} \Phi_1 + C_{12} \Phi_2$$

$$Q_2 = C_{21} \Phi_1 + C_{22} \Phi_2$$

$$Q_1 = C_{11}(\Phi_1 - \Phi_2)$$

Algebra

$$C_{11} = \frac{Q_1}{\Phi_1 - \Phi_2}$$

$$C = \frac{Q}{\Delta\Phi}$$

$\Delta\Phi$ is the difference in electrostatic potential which is commonly referred to as voltage (V), so

$$C = \frac{Q}{\Delta V}$$

3.6 Surface Charge Density

Take Gauss's Law in integral form

$$\oint_S \vec{E} \cdot n da = \frac{\int_V \rho_q(x) d^3x}{\epsilon_0}$$

Take this integral in the form of a charged sheet

$$\oint_S \vec{E} \cdot n da = \frac{\int_V \rho_{sheet}(x) d^3x}{\epsilon_0}$$

With out definition of ρ_{sheet} in terms of Dirac Delta function

$$\oint_S \vec{E} \cdot n da = \frac{\int_V \sigma \delta(n' - n'_0) d^3x}{\epsilon_0}$$

where n' is the position vector normal to the surface and n'_0 is the location on that vector that meets the surface.

The top, bottom and "thickness" are considered. The "thickness" cancels itself out because it is a loop parallel to the electric fields. The volume chosen is small enough that the surfaces are considered flat; therefore, $n = n'$ on the top side and $n = -n'$ on the bottom.

$$\oint_S \vec{E}_2 \cdot n' d^2x - \oint_S \vec{E}_1 \cdot n' d^2x = \frac{\int_V \sigma \delta(n' - n'_0) d^3x}{\epsilon_0}$$

If the area is chosen small enough then the sheet would not experience different electric fields in different locations

$$\vec{E}_2 \cdot n' \oint_S d^2x - \vec{E}_1 \cdot n' \oint_S d^2x = \frac{\int_V \sigma \delta(n' - n'_0) d^3x}{\epsilon_0}$$

The surface charge density (σ) is "distributed" so it can be taken out of integral

$$\vec{E}_2 \cdot n' \oint_S d^2x - \vec{E}_1 \cdot n' \oint_S d^2x = \frac{\sigma}{\epsilon_0} \int_V \delta(n' - n'_0) d^3x$$

$$\vec{E}_2 \cdot n' \oint_S d^2x - \vec{E}_1 \cdot n' \oint_S d^2x = \frac{\sigma}{\epsilon_0} \oint_S d^2x \int_V \delta(n' - n'_0) d^3x$$

Get ride of all $\oint_S d^2x$

$$\vec{E}_2 \cdot n' - \vec{E}_1 \cdot n' = \frac{\sigma}{\epsilon_0} \int_V \delta(n' - n'_0) d^3x$$

Take integral, but $n' = n'_0$

$$\boxed{(\vec{E}_2 - \vec{E}_1) \cdot n' = \frac{\sigma}{\epsilon_0}}$$

This equation is called "Surface Charge Density of an Arbitrary Surface. It can be used for the following

- To find the surface charge density if we know the fields
- To connect two regions at their boundary if we know the fields in both regions up to a constant
- To find the fields everywhere if we know the surface charge density and the geometry

For case 2

As seen before. Electrostatic fields are irrotational

$$\nabla \times \vec{E} = 0$$

Integrate over some arbitrary open surface

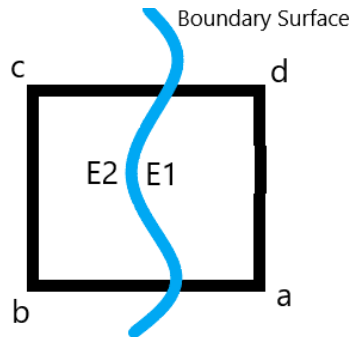
$$\oint (\nabla \times \vec{E}) \cdot n d\vec{a} = 0$$

Use Stoke's theorem to convert this to a closed line integral around the edge of the surface

$$\oint \vec{E} \cdot d\vec{x} = 0$$

The above is the integral form of the statement "Electrostatic fields are ir-rotational"

Take the integral over a rectangular closed loop **that straddles the boundary surface**.
Not a surface.



(d) Closed loop straddling the boundary surface

$$\int_a^b \vec{E}_1 \cdot d\vec{x} + \int_b^a \vec{E}_2 \cdot d\vec{x} = 0$$

Electric field is constant along length of loop

$$\vec{E}_1 \cdot \int_a^b d\vec{x} + \vec{E}_2 \cdot \int_b^a d\vec{x} = 0$$

change direction of integral to equate

$$\vec{E}_1 \cdot \int_a^b d\vec{x} - \vec{E}_2 \cdot \int_a^b d\vec{x} = 0$$

$$E_1 = E_2$$

We will call them tangential components of electric field ($E_{1,T}$ and $E_{2,T}$)

$$\boxed{\vec{E}_1 \times \vec{n} = \vec{E}_2 \times \vec{n}}$$

3.7 Maxwell's Boundary Conditions

There should be six conditions

Surface Charge Density on an Arbitrary Surface

$$\boxed{(\vec{E}_2 - \vec{E}_1) \cdot \vec{n}' = \frac{\sigma}{\epsilon_0}}$$

The components of the electric field that are tangential to a surface are continuous

$$\boxed{\vec{E}_1 \times \vec{n} = \vec{E}_2 \times \vec{n}}$$

3.8 Greens Functions

Coulomb's Law can be transformed to include boundary conditions instead of an integral over the whole universe through the use of mathematical tools called Green functions

Greens functions are applications of the divergence theorem

$$\int_V \nabla \cdot A d^3x = \int_S A \cdot da$$

A is an arbitrary vector field. Define $A = \phi \nabla \Psi$, where ϕ and Ψ are arbitrary scalars

$$\int_V \nabla \cdot \phi \nabla \Psi d^3x = \int_S \phi \nabla \Psi \cdot da$$

Use property of dot product

$$\int_V \nabla \phi \cdot \nabla \Psi + \phi \nabla^2 \Psi d^3x = \int_S \phi \frac{d\Psi}{dn} da$$

Interchange scalars and subtract the two equations

$$\int_V \nabla \phi \cdot \nabla \Psi - \nabla \Psi \cdot \nabla \phi + \phi \nabla^2 \Psi - \Psi \nabla^2 \phi d^3x = \int_S \phi \frac{d\Psi}{dn} - \Psi \frac{d\phi}{dn} da$$

Simplify

$$\int_V \phi \nabla^2 \Psi - \Psi \nabla^2 \phi d^3x = \int_S \phi \frac{d\Psi}{dn} - \Psi \frac{d\phi}{dn} da$$

Assign $\phi = \Phi$ and $\Psi = G(x, x') = \frac{1}{|x-x'|} + F(x, x')$

$$\int_V \Phi \nabla^2 G(x, x') - G(x, x') \nabla^2 \Phi d^3x = \int_S \Phi \frac{dG(x, x')}{dn} - G(x, x') \frac{d\Phi}{dn} da$$

Require that $\nabla^2 F(x, x') = 0$

$$\nabla^2 G(x, x') = \nabla^2 \frac{1}{|x-x'|}$$

Use Identity $\nabla^2 \frac{1}{|x-x'|} = -4\pi\delta(x - x')$

$$\int_V \Phi(-4\pi\delta(x - x')) - G(x, x') \nabla^2 \Phi d^3x = \int_S \Phi \frac{dG(x, x')}{dn} - G(x, x') \frac{d\Phi}{dn} da$$

Use Poisson's Equation and Evaluate the integral

$$-4\pi\Phi(x) + \frac{\int \rho(x') G d^3x}{\epsilon_0} = \int_S \Phi \frac{dG(x, x')}{dn} - G(x, x') \frac{d\Phi}{dn} da$$

Isolate Scalar potential ($\Phi(x)$)

$$\Phi(x) = \frac{\int \rho(x') G d^3x}{4\pi\epsilon_0} + \frac{\int_S \Phi \frac{dG(x, x')}{dn} - G(x, x') \frac{d\Phi}{dn} da}{4\pi}$$

This is the Green function solution to the electrostatic equations

This practically converts an "integral over all space" into a "integral over a limited volume" plus a correction term.

The "Correction term" $\frac{\int \rho(x') G d^3x}{4\pi\epsilon_0}$ is the integral of the boundary conditions over the surface enclosing the volume

The Green Function (G) is the weighting parameter and it is problem dependent. We must find the right function F that makes one of the last two terms vanish.

For Dirichlet boundary condition ($\Phi(x')$ is known on the surface of integration), we assign $G = 0$

$$\Phi(x) = \frac{\int \rho(x') G d^3x}{4\pi\epsilon_0} + \frac{\int_S \Phi \frac{dG}{dn} da}{4\pi}$$

for Neumann boundary condition ($\frac{d\Phi(x')}{dn}$ is known on the surface of integration) we assign $\frac{dG}{dn} = \frac{-4\pi}{S}$ where S is the total surface area

$$\Phi(x) = \frac{\int \rho(x') G d^3x}{4\pi\epsilon_0} + \frac{\int_S G \frac{d\Phi}{dn} da}{4\pi} + \langle \Phi \rangle_S$$

if the surface is infinite than $\langle \Phi \rangle_S = 0$

We learn later how to find the right green function

3.9 Energy Density

Previously Formulated

$$W = q(\Phi_B - \Phi_A)$$

Test point charge is brought in from infinity making $\Phi_A = 0$

$$W = q(\Phi)$$

expanding Φ

$$w = qk \int \frac{\rho(x')}{|x-x'|} dx'$$

Expanding k

$$w = \frac{q}{4\pi\epsilon_0} \int \frac{\rho(x')}{|x-x'|} dx'$$

Bring multiple charges in from infinity, one at a time

$$w = \frac{q}{8\pi\epsilon_0} \int \int \frac{\rho(x)\rho(x')}{|x-x'|} dx dx'$$

Using Coulomb's Law and the Poissons Equation

$$-\frac{\epsilon_0}{2} \int \Phi \nabla^2 \Phi dx$$

Integration by parts

$$\frac{\epsilon_0}{2} \int |\nabla \Phi|^2 dx$$

previously formulated

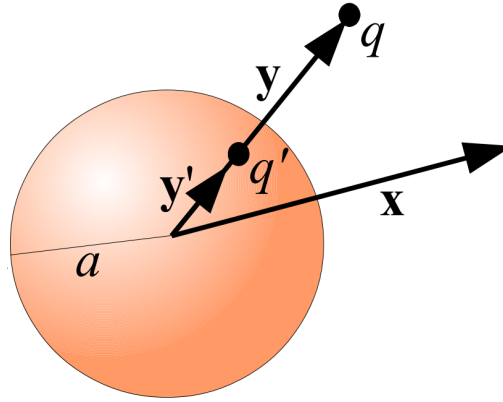
$$\frac{\epsilon_0}{2} \int |\vec{E}|^2 dx$$

$\vec{E}$ is not dependent on x

$$\frac{|\vec{E}|^2 \epsilon_0}{2}$$

3.10 Point Charge in the Presence of a Grounded Sphere

y as the distance to point charge from the origin, y' as the distance to the image charge from the origin, and a as radius of sphere as shown in figure



(e) Point charge outside of a charged sphere with a charge "image" on the inside of the sphere

Potential due to the two charges

$$\Phi(x) = \frac{1}{4\pi\epsilon_0} \frac{q}{|\vec{x} - \vec{y}|} + \frac{1}{4\pi\epsilon_0} \frac{q'}{|\vec{x} - \vec{y}'|} \quad (73)$$

$\vec{y}$ and $\vec{y}'$ point in the same direction; therefore, $\vec{y}' = y'\hat{y}$. Also substitute

using vector property ref $\vec{y} = y\hat{y}$ and $\vec{x} = x\hat{x}$

$$\Phi(x) = \frac{1}{4\pi\epsilon_0} \frac{q}{|x\hat{x} - y\hat{y}|} + \frac{1}{4\pi\epsilon_0} \frac{q'}{|x\hat{x} - y'\hat{y}|} \quad (74)$$

Apply Boundary $x = a$

$$\Phi(x) = \frac{1}{4\pi\epsilon_0} \frac{q}{|a\hat{x} - y\hat{y}|} + \frac{1}{4\pi\epsilon_0} \frac{q'}{|a\hat{x} - y'\hat{y}|} \quad (75)$$

Simplify

$$-\frac{q}{|a\hat{x} - y\hat{y}|} = \frac{q'}{|a\hat{x} - y'\hat{y}|} \quad (76)$$

The previous equation is true for all $\hat{x}$ so we can chose $\hat{x}$ that simplifies the problem

First, lets chose the $\hat{x}$ and $\hat{y}$ to be in the same direction, i.e. $\hat{x} = \hat{y}$

$$-\frac{q}{|a\hat{y} - y\hat{y}|} = \frac{q'}{|a\hat{y} - y'\hat{y}|} \quad (77)$$

Simplify

$$-\frac{q}{y - a} = \frac{q'}{a - y'} \quad (78)$$

Now we chose $\hat{x}$ to be perpendicular to $\hat{y}$ using property ref

$$|c_1\hat{r}_1 - c_2\hat{r}_2| = \sqrt{c_1^2 + c_2^2 - 2c_1c_2\hat{r}_1 \cdot \hat{r}_2}$$

$$-\frac{q}{\sqrt{a^2 + y^2 - 2ay\hat{x} \cdot \hat{y}}} = \frac{q'}{\sqrt{a^2 + y'^2 - 2ay'\hat{x} \cdot \hat{y}'}} \quad (79)$$

Simplify

$$-\frac{q}{\sqrt{a^2 + y^2}} = \frac{q'}{\sqrt{a^2 + y'^2}} \quad (80)$$

Isolate q' in equation 78

$$q' = \frac{q(y' - a)}{y - a} \quad (81)$$

Substitue equation 81 into equation 80

$$-\frac{q}{\sqrt{a^2 + y^2}} = \frac{\frac{q(y'-a)}{y-a}}{\sqrt{a^2 + y'^2}} \quad (82)$$

Get into quadratic form

$$y'^2 + \frac{a^2 + y^2}{-y}y' + a^2 = 0 \quad (83)$$

Solve quadratic equation for y'

$$y' = \frac{a^2}{y} \quad (84)$$

Substitute equation 84 into equation 81

$$q' = \frac{q((\frac{a^2}{y}) - a)}{y - a} \quad (85)$$

Simplify

$$q' = -\frac{a}{y}q \quad (86)$$

Substitute equation 86 and 84 into 73

$$\Phi(x) = \frac{1}{4\pi\epsilon_0} \frac{q}{|\vec{x} - \vec{y}|} + \frac{1}{4\pi\epsilon_0} \frac{-\frac{a}{y}q}{|\vec{x} - \frac{a^2}{y}\vec{y}|} \quad (87)$$

Simplify

$$\Phi(x) = \frac{q}{4\pi\epsilon_0} \left(\frac{1}{|\vec{x} - \vec{y}|} - \frac{1}{|\frac{y}{a}\vec{x} - \frac{a}{y}\vec{y}|} \right) \quad (88)$$

Convert to spherical coordinates

$$\Phi(r) = \frac{q}{4\pi\epsilon_0} \left(\frac{1}{\sqrt{r^2 + y^2 - 2ry\cos(\theta)}} - \frac{1}{\sqrt{\frac{y^2}{a^2}r^2 - 2ry\cos(\theta)}} \right) \quad (89)$$

We can use equation 89 to find the induced surface charge density on the sphere.

using equation ?? a.k.a the "Surface Charge Density of an Arbitrary Surface

$$(\vec{E}_2 - \vec{E}_1) \cdot \vec{n}' = \frac{\sigma}{\epsilon_0}$$

E_1 represents the electric field within the conductor, which we know, $E_1 = 0$ from equation ??

$$\vec{E}_2 \cdot \vec{n} = \frac{\sigma}{\epsilon_0} \quad (90)$$

The electric field at the surface of the conductor are parallel to the normal vector, as seen in equation ??

$$E_2 = \frac{\sigma}{\epsilon_0} \quad (91)$$

Isolate charge density (σ)

$$\sigma = -E_2\epsilon_0 \quad (92)$$

Substitute equation ??

$$\sigma = -\epsilon_0 \frac{d\Phi}{dr} \quad (93)$$

substitute equation ??

$$\sigma = -\epsilon_0 \frac{d}{dr} \left(\frac{q}{4\pi\epsilon_0} \left(\frac{1}{\sqrt{r^2 + y^2 - 2ry\cos(\theta)}} - \frac{1}{\sqrt{\frac{y^2}{a^2}r^2 - 2ry\cos(\theta)}} \right) \right) \quad (94)$$

Evaluate derivative

$$\sigma = -\frac{q}{4\pi a^2} \left(\frac{a}{y} \right) \frac{1 - \frac{a^2}{y^2}}{\left(1 + \frac{a^2}{y^2} - 2\frac{a}{y}\cos(\theta) \right)^{\frac{3}{2}}} \quad (95)$$

3.11 Point Charge in the Presence of Charged, Insulated, Conducting Sphere

Consider a sphere with total charge Q in the presence of a point charge q . There are three charges, the real

1. The real charge q at y
2. The image charge q' at y'
3. The image charge $Q - q' = Qq\frac{a}{y}$ at the origin

Using the solution for the potential from equation 88, but including the new conducting sphere factor

$$\Phi(x) = \frac{1}{4\pi\epsilon_0} \left(\frac{q}{|\vec{x} - \vec{y}|} - \frac{q}{\left| \frac{y}{a}\vec{x} - \frac{a}{y}\vec{y} \right|} + \frac{Q + q\frac{a}{y}}{|\vec{x}|} \right) \quad (96)$$

Convert to Spherical Coordinates and take out q

$$\Phi(r) = \frac{q}{4\pi\epsilon_0} \left(\frac{1}{\sqrt{r^2 + y^2 - 2ry\cos(\theta)}} - \frac{1}{\sqrt{\frac{y^2}{a^2}r^2 - 2ry\cos(\theta)}} + \frac{\frac{Q}{a} + \frac{a}{y}}{r} \right) \quad (97)$$

Use equation 93 and solve

$$\sigma = -\frac{q}{4\pi a^2} \left(\frac{a}{y} \right) \frac{1 - \frac{a^2}{y^2}}{\left(1 + \frac{a^2}{y^2} - 2\frac{a}{y}\cos(\theta) \right)^{\frac{3}{2}}} + \frac{1}{4\pi a^2} \left(Q + \frac{a}{y}q \right) \quad (98)$$

The first term is the induced surface charge found previously, and the second term is the remaining charge spready uniformly over the area of the sphere

$$\sigma = \sigma_{induced} + \frac{|Q - q'|}{A_{sphere}} \quad (99)$$

Using equation ??

$$\vec{F} = \frac{q}{4\pi\epsilon_0} \sum_i q_i \frac{(x - x_i)}{|x - x_i|^3}$$

Include additional term for charged sphere

$$\vec{F} = \frac{q}{4\pi\epsilon_0} \left(q' \frac{(y - y')}{|y - y'|^3} + (Q - q') \frac{y}{|y|^3} \right) \quad (100)$$

Consolidate

$$\vec{F} = \frac{1}{4\pi\epsilon_0} \frac{q}{y^2} \left(Q - \frac{qa^3(2y^2 - a^2)}{y(y^2 - A^2)^2} \right) \hat{y} \quad (101)$$

The potential due to a unit source and its image that satisfies homogenous boundary conditions is the Green function for Dirichlet boundary conditions (Reference equation ??)

$$G(x, x') = \frac{1}{|x - x'|} - \frac{1}{\left| \frac{x'}{a} \vec{x} - \frac{a}{x'} \vec{x}' \right|}$$

Convert to spherical coordinates using γ where
 $\cos(\gamma) = \cos(\theta)\cos(\theta') + \sin(\theta)\sin(\theta')\cos(\phi - \phi')$

$$G(x, x') = \frac{1}{\sqrt{x^2 + x'^2 - 2xx'\cos(\gamma)}} - \frac{1}{\sqrt{\frac{x'^2}{a^2} + a^2 - 2xx'\cos(\gamma)}} \quad (102)$$

This is the "Spherical Green Function for Dirichlet boundary's" and it is for both the inside and outside the sphere.

Keep in mind that $r = x$, $r' = x'$, and a is the radius of the sphere

For Greens function to be useful, we must know its derivative normal to the surface (i.e. external; therefore $n' = -x'$)

$$\left(\frac{dG}{dn'}\right)_{x'=a} = -\frac{(x^2 - a^2)}{a(x^2 + a^2 - 2xacos(\gamma))^{\frac{3}{2}}} \quad (103)$$

This is proportional to the surface charge density induced by a unit charge

For a problem inside the sphere the sign just flips

For example, using equation ??

$$\Psi(x) = -\frac{1}{4\pi} \oint (\Psi \frac{dG_D}{dn'}) da'$$

Substitute equation ??

$$\Psi(x) = \frac{1}{4\pi} \oint (\Psi \frac{(x^2 - a^2)}{a(x^2 + a^2 - 2xacos(\gamma))^{\frac{3}{2}}}) da' \quad (104)$$

Separate integral into parts

$$\Psi(x) = \frac{1}{4\pi} \int_0^{2\pi} \int_0^\pi (\Psi(a, \theta', \phi') a \frac{(x^2 - a^2)}{a(x^2 + a^2 - 2xacos(\gamma))^{\frac{3}{2}}}) \sin(\theta') d\theta' d\phi' \quad (105)$$

Conducting Sphere with Hemispheres at different potentials

3.12 Orthogonal Functions and Expansions and Fourier - Math

In the interval (a,b) of the variable x, a set of real or complex functions ($U_n(x)$) where $n = 1, 2$ are **orthogonal** when $m \neq n$ if:

$$\int_a^b U_n^*(x) U_m(x) dx = 0 \quad (106)$$

When $m = n$ and the functions are normalized they are **orthonormal**

$$\int_a^b U_n^*(x)U_m(x)dx = \delta_{nm} \quad (107)$$

An arbitrary, integrable function ($f(x)$) can be expanded in a series of the orthonormal functions ($U_n(x)$) per

$$f(x) = \sum_{n=2}^N a_n U_n(x) \quad (108)$$

To get unknown coefficients we would need to normalize
Multiply both sides by $U_m^*(x)$

$$f(x)U_m^*(x) = \sum_{n=2}^N a_n U_n(x)U_m^*(x) \quad (109)$$

Integrate

$$\int_a^b f(x)U_m^*(x)dx = \sum_{n=2}^N a_n \int_a^b U_n(x)U_m^*(x)dx \quad (110)$$

Substitute equation ??

$$\int_a^b f(x)U_m^*(x)dx = \sum_{n=2}^N a_n \delta_{nm} \quad (111)$$

$$\int_a^b f(x)U_m^*(x)dx = a_m \quad (112)$$

Since the initial assumption for equation ?? is that $m = n$, swap subscripts
subscripts to match initial function definition

$$\int_a^b f(x)U_m^*(x)dx = a_n \quad (113)$$

Place into function expansion (Reference equation ??) to get "Finite Series Expansion". If the functions form a complete set, which all useful physical function sets are, then this becomes a more accurate representation of the function $f(x)$, as more terms in the series are kept.

More accurately is the "Infinite Series"

$$f(x) = \sum_{n=2}^{\infty} a_n U_n(x) \quad (114)$$

Or maybe you want an "Infinite Continuous Expansion" where the orthogonal functions become a continuum of functions, the index variable n becomes a continuous variable k .

$$\int_{-\infty}^{\infty} U_k^*(x)U_{k'}(x)dx = \delta_{k-k'} \quad (115)$$

$$f(x) = \int_{-\infty}^{\infty} A(k)U_k(x)dk \quad (116)$$

$$\int_{-\infty}^{\infty} f(x)U_n^*(x)dx = A(k) \quad (117)$$

The most common orthogonal functions are sines and cosines, constituting Fourier Series

$$f(x) = \sum_{n=0}^{\infty} A_n \cos(k_n x) + B_n \sin(k_n x) \quad (118)$$

For this series to be valid it must be periodic such that $f(-\frac{a}{2}) - f(\frac{a}{2}) = 0$; therefore,

$$\sum_{n=0}^{\infty} A_n (\cos(k_n(-\frac{a}{2})) - \cos(k_n \frac{a}{2})) + B_n (\sin(k_n(-\frac{a}{2})) - \sin(k_n \frac{a}{2})) = 0 \quad (119)$$

This must be true independent of A_n and B_n , so that the coefficients must be zero

$$(\sin(k_n(-\frac{a}{2})) - \sin(k_n \frac{a}{2})) = 0 \quad (120)$$

If one is zero, both are zero.

$$\sin(k_n \frac{a}{2}) = 0 \quad (121)$$

for $\sin(x) = 0$, $x = n\pi$; therefore,

$$k_n \frac{a}{2} = n\pi \quad (122)$$

Isolate k_n

$$k_n = \frac{2\pi n}{a} \quad (123)$$

Substitute equation ?? into equation ??

$$f(x) = \sum_{n=0}^{\infty} A_n \cos(\frac{2\pi n x}{a}) + B_n \sin(\frac{2\pi n x}{a}) \quad (124)$$

To find coefficients

multiply both sides by $\cos(\frac{2\pi m x}{a})$:

$$f(x) \cos(\frac{2\pi m x}{a}) = \sum_{n=0}^{\infty} A_n \cos(\frac{2\pi n x}{a}) \cos(\frac{2\pi m x}{a}) + B_n \sin(\frac{2\pi n x}{a}) \cos(\frac{2\pi m x}{a}) \quad (125)$$

integrate over $(-\frac{a}{2}, \frac{a}{2})$

$$\int_{-\frac{a}{2}}^{\frac{a}{2}} f(x) \cos(\frac{2\pi m x}{a}) dx = \int_{-\frac{a}{2}}^{\frac{a}{2}} \sum_{n=0}^{\infty} A_n \cos(\frac{2\pi n x}{a}) \cos(\frac{2\pi m x}{a}) + B_n \sin(\frac{2\pi n x}{a}) \cos(\frac{2\pi m x}{a}) dx \quad (126)$$

partially Evaluate Integral, since,

$$\int \cos(x)\sin(x) = \frac{\sin^2(x)}{2} + C$$

$$\sin^2(\pi) = 0$$

The second term in equation ?? can be canceled
the first term in equation ?? is zero except when $m = n$ due to orthogonality

$$A_n = \frac{2}{a} \int_{-\frac{a}{2}}^{\frac{a}{2}} f(x) \cos\left(\frac{2\pi mx}{a}\right) dx \quad (127)$$

To find B_n , multiply both sides of equation ?? and integrate

$$B_n = \frac{2}{a} \int_{-\frac{a}{2}}^{\frac{a}{2}} f(x) \sin\left(\frac{2\pi mx}{a}\right) dx \quad (128)$$

More useful version of Fourier series is in terms of complex exponentials
because the summation index spans negative to positive infinity

$$f(x) = \frac{1}{\sqrt{a}} \sum_{n=-\infty}^{\infty} A_n e^{i\left(\frac{2\pi nx}{a}\right)} \quad (129)$$

where

$$A_n = \frac{1}{\sqrt{a}} \int_{-\frac{a}{2}}^{\frac{a}{2}} f(x') e^{-i\left(\frac{2\pi nx'}{a}\right)} dx' \quad (130)$$

If the interval becomes infinite ($a \rightarrow \infty$)

$$f(x) = \int_{-\infty}^{\infty} A(k) e^{ikx} dk \quad (131)$$

To find coefficient A_n , multiply both sides by $e^{-ik'x}$ and integrate

$$\int_{-\infty}^{\infty} f(x) e^{-ik'x} dx = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} A(k) e^{ikx} e^{-ik'x} dx dk \quad (132)$$

Using Orthogonality of complex exponential

$$\int_{-\infty}^{\infty} e^{-ik'x} dx = 2\pi \delta(k - k') \quad (133)$$

Substitute

$$A(k) = \frac{1}{2\pi} \int_{-\infty}^{\infty} f(x) e^{-ikx} dx \quad (134)$$

$$A(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-ikx} dx \quad (135)$$

$$f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} A(k) e^{ikx} dk \quad (136)$$

”Fourier Integral”

3.13 Separation of Variables: Laplace Equation in Rectangular Coordinates

The Laplace equation (Reference equation ??)

$$\nabla^2 \Phi = 0$$

Expand ∇ into rectangular coordinates

$$ay o \tag{137}$$

3.14 Laplace Equation in Spherical Coordinates

The definition of the Laplace equation

$$\nabla^2 \Phi = 0 \tag{138}$$

The definition of ∇ in spherical coordinates

$$\nabla = \hat{e}_1 \frac{\partial}{\partial r} + \hat{e}_2 \frac{1}{\theta} \frac{\partial}{\partial \theta} + \hat{e}_3 \frac{1}{r \sin(\theta)} \frac{\partial}{\partial \phi} \tag{139}$$

Where r is the radial distance from the origin, θ is the polar angle with the z-axis, ϕ is the azimuthal angle in the x-y plan relative to the x-axis

Combine the two

$$\frac{1}{r} \frac{\partial^2}{\partial r^2} (r\Phi) + \frac{1}{r^2 \sin(\theta)} \frac{\partial}{\partial \theta} (\sin(\theta) \frac{\partial \Phi}{\partial \theta}) + \frac{1}{r^2 \sin^2(\theta)} \frac{\partial^2 \Phi}{\partial \phi^2} = 0 \tag{140}$$

Use method of separation of variables by trying a solution of the form

$$\Phi(r, \theta, \phi) = \frac{R(r)}{r} P(\theta) Q(\phi) \tag{141}$$

Substitute equation 141 into equation 140

$$\frac{1}{r} \frac{\partial^2}{\partial r^2} (r \frac{R(r)}{r} P(\theta) Q(\phi)) + \frac{1}{r^2 \sin(\theta)} \frac{\partial}{\partial \theta} (\sin(\theta) \frac{\partial}{\partial \theta} (\frac{R(r)}{r} P(\theta) Q(\phi))) + \frac{1}{r^2 \sin^2(\theta)} \frac{\partial^2}{\partial \phi^2} (\frac{R(r)}{r} P(\theta) Q(\phi)) = 0 \tag{142}$$

Multiply by $\frac{r^3 \sin^2(\theta)}{R(r) P(\theta) Q(\phi)}$

$$\frac{r^2 \sin^2(\theta)}{R(r)} \frac{d^2 R(r)}{dr^2} + \frac{\sin(\theta)}{P(\theta)} \frac{d}{d\theta} (\sin(\theta) \frac{dP(\theta)}{d\theta}) + \frac{1}{Q(\phi)} \frac{d^2 Q(\phi)}{d\phi^2} = 0 \tag{143}$$

The last term is solely dependent on ϕ so it can be equal to a constant

$$\frac{r^2 \sin^2(\theta)}{R(r)} \frac{d^2 R(r)}{dr^2} + \frac{\sin(\theta)}{P(\theta)} \frac{d}{d\theta} (\sin(\theta) \frac{dP(\theta)}{d\theta}) - m^2 = 0 \tag{144}$$

Where

$$\frac{1}{Q(\phi)} \frac{d^2 Q(\phi)}{d\phi^2} = -m^2 \tag{145}$$

Rearrange to get common solution

$$\frac{d^2 Q(\phi)}{d\phi^2} = -m^2 Q(\phi) \quad (146)$$

Common solutions if $m \neq 0$

$$Q(\phi) = A_m e^{im\phi} + B_m e^{-im\phi} \quad (147)$$

Common solutions if $m = 0$

$$Q(\phi) = A_m + B_m \phi \quad (148)$$

Back to Original equation 144, multiply $\sin^2(\theta)$

$$\frac{r^2}{R(r)} \frac{d^2 R(r)}{dr^2} + \frac{1}{\sin(\theta)P(\theta)} \frac{d}{d\theta} \left(\sin(\theta) \frac{dP(\theta)}{d\theta} \right) - \frac{m^2}{\sin^2 \theta} = 0 \quad (149)$$

First and Last terms are independent so they can be set to constants

$$\frac{r^2}{R(r)} \frac{d^2 R(r)}{dr^2} - l(l+1) = 0 \quad (150)$$

Where

$$-l(l+1) = \frac{1}{\sin(\theta)P(\theta)} \frac{d}{d\theta} \left(\sin(\theta) \frac{dP(\theta)}{d\theta} \right) - \frac{m^2}{\sin^2 \theta} \quad (151)$$

Put equation 150 into more intuitive forms

$$\frac{d^2 R(r)}{dr^2} = l(l+1) \frac{R(r)}{r^2} \quad (152)$$

Put equation 151 into more intuitive forms

$$\frac{d}{d\theta} \left(\sin(\theta) \frac{dP(\theta)}{d\theta} \right) + (l(l+1) - \frac{m^2}{\sin^2 \theta}) \sin(\theta) P(\theta) = 0 \quad (153)$$

Reduce equation 153 to simpler form by substituting $x = \cos(\theta)$ and $\frac{d}{d\theta} = -\sqrt{1-x^2} \frac{d}{dx}$

$$\frac{d}{dx} \left((1-x^2) \frac{dP(X)}{dx} \right) + (l(l+1) - \frac{m^2}{1-x^2}) P(x) = 0 \quad (154)$$

Equation 152 by substituting $R(r) = r^\alpha$

$$\alpha(\alpha-1)r^{\alpha-2} = (l+1)r^{\alpha-2} \quad (155)$$

$$\alpha^2 \alpha - 1 - l(l+1) = 0 \quad (156)$$

$$\alpha = l+1$$

$$\alpha = -l$$

General solution is

$$R(r) = A_l r^{l+1} + B_l r^{-1} \quad (157)$$

General General Solution

$$\Phi(r, \theta, \phi) = \sum_m \sum_l (A_l r^l + B_l r^{-l-1}) (A_m d^{im\phi} + B_m e^{-im\phi} P_l^m(\cos(\theta))) \quad (158)$$

Ordinary Legendre Polynomials (m=0)

using $m = 0$ and equation 154

$$\frac{d}{dx} \left((1-x^2) \frac{dP(X)}{dx} \right) + l(l+1)P(x) = 0 \quad (159)$$

Guess ("Try") power series solution

$$P(x) = \sum_{j=0}^{\infty} a_j x^{j+\alpha} \quad (160)$$

$$\sum_{j=0}^{\infty} (j+\alpha)(j+\alpha-1)a_j x^{j+\alpha-2} - (j+\alpha)(j+\alpha+1)a_j x^{j+\alpha} + l(l+1)a_j x^{j+\alpha} = 0 \quad (161)$$

Consolidate

$$\sum_{j=0}^{\infty} (j+\alpha)(j+\alpha-1)a_j x^{j+\alpha-2} - \sum_{j=0}^{\infty} ((j+\alpha)(j+\alpha+1) + l(l+1))a_j x^{j+\alpha} = 0 \quad (162)$$

Remove first two powers of x then combine

$$(\alpha)(\alpha-1)a_0 x^{\alpha-2} + (1+\alpha)(\alpha)a_1 x^{\alpha-2} - \sum_{j=0}^{\infty} ((j+2+\alpha)(j+\alpha+1)a_{j+2} - ((j+\alpha)(j+\alpha+1) + l(l+1))a_j)x^{j+\alpha} = 0 \quad (163)$$

This must hold for all values of x so

$$(\alpha)(\alpha-1)a_0 = 0 \quad (164)$$

$$(1+\alpha)(\alpha)a_1 = 0 \quad (165)$$

Substitute equation ?? into ??

$$(j+\alpha+2)(j+\alpha+1)a_{j+2} - ((j+\alpha)(j+\alpha+1) + l(l+1))a_j = 0 \quad (166)$$

The third equation is solved to yield the recurrence relation which gives all the remaining terms from the last one. Rerrange equation ??

$$a_{j+2} = \frac{(j+\alpha)(j+\alpha+1) - l(l+1)}{(j+\alpha+2)(j+\alpha+1)} a_j \quad (167)$$

$a_1 = 0$; therefore, $a_{odd} = 0$

$$P_l(x) = \sum_{j=0, even}^{\infty} a_j x^{j+\alpha} \quad (168)$$

Convergest at $x = 1$

$$\frac{(j_{max} + \alpha)(j_{max} + \alpha + 1) - l(l + 1)}{(j_{max} + \alpha + 2)(j_{max} + \alpha + 1)} = 0 \quad (169)$$

If $\alpha = 0$

$$j_{max}(j_{max} + 1) - l(l + 1) = 0 \quad (170)$$

Therefore

$$j_{max} = l \quad (171)$$

Substitute equation ?? with ??

$$P_l(x) = \sum_{j=0, even}^l a_j x^j \quad (172)$$

Therefore

$$P_l(x) = \sum_{j=0}^{\frac{l}{2}} a_{2j} x^{2j} \quad (173)$$

Rewrite sums over all integers by doubling indices. If l is even

$$a_{j+2} = j \frac{(j + 1) - l(l + 1)}{(j + 2)(j + 1)} a_j \quad (174)$$

If $\alpha = 1$

$$(j_{max} + 2)(j_{max} + 1) - l(l + 1) = 0 \quad (175)$$

Therefore

$$j_{max} = l - 1 \quad (176)$$

Substitute equation ?? with ??

$$P_l(x) = \sum_{j=0, even}^{l-1} a_j x^{j+1} \quad (177)$$

Rewrite sums over all integers by doubling indices. If l is odd

$$a_{j+2} = \frac{(j + 1)(j + 2) - l(l + 1)}{(j + 3)(j + 2)} a_j \quad (178)$$

Legendre polynomials have the following mathematical properties

Odd or even symmetry about origin $P_l(-x) = (-1)^l P_l(x)$

Rodrigues' Formula $P_l(x) = \frac{1}{2^l l!} \frac{d^l}{dx^l} (x^2 - 1)^l$

Recurrence relations: $P_{l+1}(x) = \frac{2l+1}{l+1} P_l(x) - \frac{l}{l+1} P_{l-1}(x)$

and $P_l(x) = \frac{1}{2l+1} \frac{d}{dx} (P_{l+1}(x) - P_{l-1}(x))$

Legendre polynomials are not defined for negative l

Orthogonality condition $\int_{-1}^1 P_{l'}(x) P_l(x) dx = \frac{2}{2l+1} \delta_{l'l}$

Orthogonality condition (Polar angle) $\int_0^\pi P_{l'}(\cos(\theta)) P_l(\cos(\theta)) \sin(\theta) d\theta = \frac{2}{2l+1} \delta_{l'l}$ only works for these integral limits

4 Notes

- Classical electrodynamics is a macroscopic theory.
- Classical electrodynamics contains electric charge, but does not contain point charges (Electrons)
- Each Dirac Delta function fixes a dimension
- Zero width objects are represented with dirac delta functions
- Units of Dirac Delta function are inverse of the units of it's operand
- L. Schwartz's theory of distributions is a comprehensive rigorous mathematical approach to delta functions and their manipulations.*
- Gauss's law holds for Newtonian gravitational force fields,
- Gauss's law depends on inverse square law for the force between charges, central nature of the force, and linear superposition of the effects of different charges.
- Mathematically, we could have chosen the sign in $\vec{E} = \nabla\Phi$. The negative gradient operator points downhill in the scalar function landscape. Balls roll downhill, electric charges feel a force downhill.
- Coulomb's electrostatic potential, analogous to the classical gravitational potential, has no physical meaning in a macroscopic system such as electrodynamics, but "seems to dictate that the potentials have physical meaning" on quantum ranges ??. Differences in scalar electrostatic potential (Voltage) can be related to things with meaning.
- Electrostatic scalar potential is always continuous (except across line charges and dipole charge layers, but these are nonphysical Idealizations. A discontinuity would lead to an infinite slope. Continuity can be used as a boundary condition for the potentials but not the fields. The scalar potential is a scalar field that exists at every point in space while the potential energy is a single number attached to a specific object that is experiencing the scalar potential at a certain point in space.